lab1

August 5, 2024

Question 1

[1]:
```
def BreakDown():
    input = input("Enter a string: ")
    output = []
    for i in input:
        output.append(i)
    print(output)
BreakDown()
```

```
['a', 'b', 'c']
```

Question 2

[5]:
```
def convert_to_string(output):
    result = ""
    for char in output:
        result += char
    print(result)
convert_to_string(['a', 'b', 'c'])
```

```
abc
```

Question 3

[6]:
```
import random
def RandomNumber(n):
    randlist = []
    for i in range(n):
        randlist.append(random.randint(1, 100))
    print(randlist)
RandomNumber(5)
```

```
[72, 74, 86, 10, 20]
```

Question 4

[3]:
```
def sortlist():
    list1 = [1, 3, 5, 7, 9]
    list1.sort(reverse=True)
```

```python
    print(list1)
sortlist()
```

```
[9, 7, 5, 3, 1]
```

Question 5

[4]:
```python
def frequency():
    list1 = [1, 1, 3, 2, 3, 2, 3, 2, 2]
    freq = {}
    for item in list1:
        if item in freq:
            freq[item] += 1
        else:
            freq[item] = 1
    print(freq)
frequency()
```

```
{1: 2, 3: 3, 2: 4}
```

Question 6

[5]:
```python
def unique():
    list1 = [1, 1, 3, 2, 3, 2, 3, 2, 2]
    unique_list = []
    set1 = set(list1)
    for i in set1:
        unique_list.append(i)
    print(unique_list)
unique()
```

```
[1, 2, 3]
```

Question 7

[6]:
```python
def first_repeating():
    list1 = [1,2,3,4,5,1,2]
    seen = set()
    for element in list1:
        if element in seen:
            return element
        seen.add(element)
    return None
print(first_repeating())
```

```
1
```

Question 8

[8]:
```python
def square_n_cube(n):
    dict1 = {}
    for i in range(0, n+1):
        dict1[i] = (i**2, i**3)
    print(dict1)
square_n_cube(3)
```

```
{0: (0, 0), 1: (1, 1), 2: (4, 8), 3: (9, 27)}
```

Question 9

[9]:
```python
def tupling():
    list1 = [1,2,3,4]
    list2 = ['a', 'b', 'c', 'd']
    result = []
    for i in range(len(list1)):
        result.append((list1[i], list2[i]))
    print(result)
tupling()
```

```
[(1, 'a'), (2, 'b'), (3, 'c'), (4, 'd')]
```

Question 10

[11]:
```python
def list_comprehension(n):
    squares = [i**2 for i in range(n+1)]
    return squares

print(list_comprehension(5))
```

```
[0, 1, 4, 9, 16, 25]
```

Question 11

[12]:
```python
def dectionary_comprehension(n):
    squares = {i: i**2 for i in range(n+1)}
    return squares
print(dectionary_comprehension(5))
```

```
{0: 0, 1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

Question 12

[22]:
```python
class List:
    def __init__(self, values):
        self.values = values

    def apply(self, func):
        try:
            result = list(map(func, self.values))
```

```
            return result
        except Exception as e:
            raise Exception("An error occurred while applying the function: " +
 ↪str(e))

list1 = List([1, 2, 3, 4, 5])
try:
    result = list1.apply(lambda x: x**2)
    print(result)
except Exception as e:
    print(e)
```

```
[1, 4, 9, 16, 25]
```

Question 13

[17]:
```
def upper_list():
    list1 = ['aa','bb','cd','e']
    upper = list(map(str.upper, list1))
    print(upper)
upper_list()
```

```
['AA', 'BB', 'CD', 'E']
```

Question 14

[18]:
```
from functools import reduce
def product():
    numbers = [1, 2, 3, 4, 5]
    product = reduce(lambda x, y: x * y, numbers)
    print(product)
product()
```

```
120
```